

# Daily Geometry Drill #1 — Answers & Investigations

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| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

*New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.*

## Section A Rapid Recognition

1. Find the gradient of the line  $6x - 3y + 15 = 0$ .

**Answer:** 2

**Method:** Rearrange to gradient–intercept form:  $3y = 6x + 15 \Rightarrow y = 2x + 5$ , so  $m = 2$ .

**Investigate further:** Implicit form  $ax + by + c = 0$  has gradient  $-a/b$ ; here  $-\frac{6}{3} = 2$ . **If the line is instead written  $3y - 6x - 15 = 0$  (the same line), is the gradient still 2?** It must be — check the  $-a/b$  rule still agrees.

2. Find the gradient of the line through  $(1, 2)$  and  $(5, 14)$ .

**Answer:** 3

**Method:**  $m = \frac{14-2}{5-1} = \frac{12}{4} = 3$ .

**Investigate further:** Check with  $(5 - 1, 14 - 2)$  swapped:  $\frac{2-14}{1-5} = \frac{-12}{-4} = 3$  ✓ — the sign always cancels. **If the line through  $(1, 2)$  and  $(5, 14)$  is moved up by 3 (every  $y$  increased by 3), does its gradient change?** No — a vertical translation keeps the ratio of rises fixed.

3. A line of gradient 4 is perpendicular to a line of gradient  $m$ . Write down  $m$ .

**Answer:**  $-\frac{1}{4}$

**Method:** Perpendicular gradients are negative reciprocals:  $m = -\frac{1}{4}$ , since  $4 \cdot (-\frac{1}{4}) = -1$ .

**Investigate further:** The product test  $m_1 m_2 = -1$  is the fastest check. **Is the line  $y = 4x$  parallel, perpendicular, or neither, to  $y = -\frac{1}{4}x + 12$ ?** The constant term is a distraction — gradients alone decide.

4. Find the distance between  $(2, 1)$  and  $(5, 5)$ .

**Answer:** 5

**Method:**  $d = \sqrt{(5-2)^2 + (5-1)^2} = \sqrt{9+16} = \sqrt{25} = 5$ .

**Investigate further:** 3-4-5 right triangle in disguise. **The points  $(-1, 0)$  and  $(3, -3)$  are the same distance apart — can you see the Pythagorean triple before computing, by drawing the displacement vector?** The displacement  $(4, -3)$  is a reordered 3-4-5.

5. Find the midpoint of the segment joining  $(4, 8)$  and  $(10, 2)$ .

**Answer:**  $(7, 5)$

**Method:** Midpoint averages each coordinate:  $(\frac{4+10}{2}, \frac{8+2}{2}) = (7, 5)$ .

**Investigate further:** Check both halves are equal distances:  $(7-4, 8-5) = (3, 3)$  and  $(10-7, 5-2) = (3, 3)$  ✓. **The midpoint of two points is the average of the two displacement vectors. What is the midpoint of  $(4, 8)$  and  $(-6, -2)$ ?** Negative coordinates behave the same way.

6. Find the perpendicular distance from the origin to the line  $3x + 4y - 10 = 0$ .

**Answer:** 2

**Method:** Use  $\frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$  at  $(0, 0)$ :  $\frac{|-10|}{\sqrt{9+16}} = \frac{10}{5} = 2$ .

**Investigate further:** The denominator 5 is the length of the normal vector  $(3, 4)$ . **The point of the line closest to the origin is a multiple  $t(3, 4)$  of the normal — find it and confirm its distance is indeed 2.**

7. A line has  $x$ -intercept 4 and  $y$ -intercept 6. Find its gradient.

**Answer:**  $-\frac{3}{2}$

**Method:** The intercepts are the points  $(4, 0)$  and  $(0, 6)$ :  $m = \frac{6-0}{0-4} = -\frac{3}{2}$ .

**Investigate further:** Reading right-to-left: this is the negative of  $\frac{y\text{-intercept}}{x\text{-intercept}}$ . **Write the line in intercept form  $\frac{x}{4} + \frac{y}{6} = 1$  and rearrange to confirm  $m = -\frac{y\text{-intercept}}{x\text{-intercept}} = -\frac{6}{4} = -\frac{3}{2}$  — the intercept form makes the gradient transparent.**

8. Find the area of the triangle with vertices  $(0, 0)$ ,  $(4, 0)$  and  $(0, 3)$ .

**Answer:** 6

**Method:** Right-angled at the origin: area =  $\frac{1}{2} \cdot 4 \cdot 3 = 6$ .

**Investigate further:** Equivalently the shoelace/midpoint-vector formula  $\frac{1}{2}|(4)(3) - (0)(0)| = 6$ . **Rotate this triangle by  $90^\circ$  about the origin: does the area change, and do you get the same shoelace number?** Reflections and rotations preserve area.

9. The points  $(1, 2)$ ,  $(2, 4)$  and  $(k, 10)$  are collinear. Find  $k$ .

**Answer:** 5

**Method:** Rise per unit run is 2 (from  $(1, 2)$  to  $(2, 4)$ ); continuing to  $y = 10$  climbs 6 more, so  $k = 2 + \frac{6}{2} = 5$ .

**Investigate further:** Formal check:  $\frac{4-2}{2-1} = \frac{10-4}{k-2} \Rightarrow 2 = \frac{6}{k-2} \Rightarrow k = 5$ . **What happens to  $k$  if the points are required to be collinear but the order is  $(2, 4)$ ,  $(1, 2)$ ,  $(k, 10)$ ?** Collinearity is order-independent — all three gradients must agree.

10. Using the shoelace formula, find the area of the triangle with vertices  $(0, 0)$ ,  $(3, 4)$  and  $(6, 0)$ .

**Answer:** 12

**Method:** Shoelace:  $\frac{1}{2}|0 \cdot 4 + 3 \cdot 0 + 6 \cdot 0 - (0 \cdot 3 + 4 \cdot 6 + 0 \cdot 0)| = \frac{1}{2}|24| = 12$ .

**Investigate further:** Bonus route: base = 6, height = 4, area =  $\frac{1}{2} \cdot 6 \cdot 4 = 12$  ✓. **For a triangle with one vertex at the origin, shoelace collapses to  $\frac{1}{2}|x_1y_2 - x_2y_1|$  with the other two points — prove that this equals  $\frac{1}{2}|\det|$  of the two vectors.** That determinant is the signed area.

## Section B Manipulation Drills

1. The line  $2x - 3y + 6 = 0$  is perpendicular to which of the following?

A)  $y = -\frac{3}{2}x + 1$

B)  $y = \frac{3}{2}x - 2$

C)  $y = \frac{2}{3}x + 5$

D)  $y = -\frac{2}{3}x + 3$

**Answer:** A)  $y = -\frac{3}{2}x + 1$

**Method:** Given line has gradient  $\frac{2}{3}$  (rule  $-a/b$ ), so perpendicular gradient is  $-\frac{3}{2}$ . Only option A has gradient  $-\frac{3}{2}$ .

**Investigate further:** Each decoy makes one half of the flip-and-negate move: option B inverts without negating, option D negates without inverting, and option C is parallel to the given line. **Which option would be perpendicular to  $3x + 2y = 1$ ?**

2. What is the perpendicular distance from  $(2, 3)$  to the line  $5x + 12y - 7 = 0$ ?

A) 1

B) 2

C) 3

D) 4

**Answer:** C) 3

**Method:**  $\frac{|5(2)+12(3)-7|}{\sqrt{25+144}} = \frac{|10+36-7|}{13} = \frac{39}{13} = 3$ .

**Investigate further:** The numerator is exactly how far the point sits from the line measured along a normal whose length is 13. **Two points are at perpendicular distance 3 from  $5x + 12y - 7 = 0$  — what do they have in common?** They lie on  $5x + 12y - 7 = \pm 39$ , the two parallel satellite lines.

3. Which point lies on the perpendicular bisector of the segment joining  $(1, 2)$  and  $(5, 6)$ ?

A)  $(2, 5)$

B)  $(3, 3)$

C)  $(4, 4)$

D)  $(5, 3)$

**Answer:** A)  $(2, 5)$

**Method:** Midpoint  $(3, 4)$ ; segment gradient 1, so bisector gradient  $-1$ :  $y - 4 = -(x - 3)$ , i.e.  $x + y = 7$ .  
Check options:  $2 + 5 = 7$  ✓.

**Investigate further:** Only A satisfies  $x + y = 7$ . **A point is on the perpendicular bisector iff it is equidistant from the two endpoints — verify  $(2, 5)$  is distance  $\sqrt{16}$  from both  $(1, 2)$  and  $(5, 6)$ .** That's the defining property, worth internalising.

4. Find the area of the quadrilateral with vertices  $(0, 0)$ ,  $(4, 0)$ ,  $(5, 3)$ ,  $(1, 3)$ , in that order.

- A) 10  
B) 12  
C) 14  
D) 16

**Answer:** B) 12

**Method:** Shoelace in order:  $\frac{1}{2}|0 \cdot 0 + 4 \cdot 3 + 5 \cdot 3 + 1 \cdot 0 - (0 \cdot 4 + 0 \cdot 5 + 3 \cdot 1 + 3 \cdot 0)| = \frac{1}{2}|0 + 12 + 15 + 0 - (0 + 0 + 3 + 0)| = \frac{1}{2}|24| = 12$ .

**Investigate further:** The quadrilateral is a parallelogram (bases  $\parallel$  via position vectors  $(4, 0)$  and  $(1, 3)$ ), so area =  $|\det| = |4(3) - 0(1)| = 12$  ✓. **Reorder the vertices as  $(0, 0)$ ,  $(5, 3)$ ,  $(4, 0)$ ,  $(1, 3)$ : the same region, but the shoelace number warns you when points are listed in the wrong order.** Keep vertices in cyclic order.

5. The line through  $(1, 2)$  and  $(4, 5)$  crosses the  $y$ -axis at which value?

- A) 0  
B) 1  
C) 2  
D) 3

**Answer:** B) 1

**Method:** Gradient  $\frac{5-2}{4-1} = 1$ ;  $y = x + c$  through  $(1, 2)$  gives  $c = 1$ .

**Investigate further:** Extension: at  $x = 0$ ,  $y = 1$ . **Through the point-slope form,  $y - 2 = 1(x - 1)$ . If the line were instead the one through  $(1, 2)$  and  $(4, 8)$ , the  $y$ -intercept changes — resolve  $c$  again.** Always plug the known point in, never guess  $c$  from the picture.

6. Find the perpendicular distance from  $(0, 1)$  to the line  $6x - 8y + 2 = 0$ .

**Answer:**  $\frac{3}{5}$

**Method:**  $\frac{|6(0) - 8(1) + 2|}{\sqrt{36 + 64}} = \frac{|-6|}{10} = \frac{3}{5}$ .

**Investigate further:** Normal length  $\sqrt{36 + 64} = 10$ ; numerator 6. **The line  $6x - 8y + 2 = 0$  sits distance  $\frac{1}{5}$  from the origin (check  $\frac{|2|}{10}$ ). Since the  $y$ -axis point  $(0, 1)$  is 1 unit up from the origin, the net distance  $|1 - \frac{1}{5}| = \frac{4}{5}$  differs from  $\frac{3}{5}$  — why?** Because the normal  $(6, -8)$  is not vertical: only the projection along it counts.

7. Which of the following is NOT parallel to  $y = 3x + 1$ ?

- A)  $y = 3x + 5$   
 B)  $6x - 2y = 4$   
 C)  $3x - y = 2$   
 D)  $y = 2x + 1$

**Answer:** D)  $y = 2x + 1$

**Method:** Parallel  $\Leftrightarrow$  same gradient 3. A:  $3\checkmark$ ; B:  $6x - 2y = 4 \Rightarrow y = 3x - 2\checkmark$ ; C:  $3x - y = 2 \Rightarrow y = 3x - 2\checkmark$ ; D:  $y = 2x + 1$  has gradient 2.

**Investigate further:** Options B and C are disguises (divide by the  $y$ -coefficient first). **Show two distinct parallel lines never intersect, while any two non-parallel lines always do.** Parallelism is exactly “same slope, different intercept”.

8. The line  $y = x + 1$  meets the line through  $(0, 3)$  perpendicular to it at the point  $P$ . Find the coordinates of  $P$ .

**Answer:** ”(1, 2)”

**Method:** Through  $(0, 3)$  with gradient  $-1$ :  $y = -x + 3$ . Intersect:  $x + 1 = -x + 3 \Rightarrow x = 1$ , then  $y = 2$ .

**Investigate further:** The intersection of two perpendicular lines is the right-angle vertex of a right triangle. Verify  $P = (1, 2)$  lies on both lines:  $y = x + 1$  gives  $2 = 1 + 1 \checkmark$  and  $y = -x + 3$  gives  $2 = -1 + 3 \checkmark$ . **Through  $(0, 3)$  draw the perpendicular and reflect the point where it meets  $y = x + 1$  across that meeting point — what do you get, and how does it connect to the midpoint of  $(0, 3)$  and  $(2, 1)$ ?** Perpendicular lines meet exactly once; the reflect-and-extend trick answers the follow-up.

9. Which point is closest to the origin?

- A)  $(1, 3)$   
 B)  $(2, 2)$   
 C)  $(4, 0)$   
 D)  $(0, 5)$

**Answer:** B)  $(2, 2)$

**Method:** Distances from the origin:  $(1, 3)$  gives  $\sqrt{10}$ ,  $(2, 2)$  gives  $\sqrt{8}$ ,  $(4, 0)$  gives 4,  $(0, 5)$  gives 5; the smallest is  $\sqrt{8}$ .

**Investigate further:** Comparing squared distances avoids surds: 10, 8, 16, 25. **Among points with  $x, y$  nonnegative integers and  $x + y = 4$ , which is closest to the origin?** Both squares and sum compare against  $(2, 2)$ .

10. Using the shoelace formula, find the area of the quadrilateral with vertices  $(1, 0)$ ,  $(5, 2)$ ,  $(4, 6)$ ,  $(0, 3)$ , in that order.

**Answer:**  $\frac{33}{2}$

**Method:** Cross terms  $x_i y_{i+1} - x_{i+1} y_i$  in order:  $1 \cdot 2 - 5 \cdot 0 = 2$ ,  $5 \cdot 6 - 4 \cdot 2 = 22$ ,  $4 \cdot 3 - 0 \cdot 6 = 12$ ,  $0 \cdot 0 - 1 \cdot 3 = -3$ . Their sum is 33, so the area is  $\frac{33}{2}$ .

**Investigate further:** List the same four vertices in the order  $(1, 0), (4, 6), (5, 2), (0, 3)$ . The shoelace sum now describes a self-crossing path: what does it give, and why is it not an area?

## Section C Substitution & Structure

( $\leq 90$  s each)

*Competition-hard: hidden structure, non-routine use of the same tools. No question is a direct formula plug.*

1. The lines  $l_1 : y = 6 - 2x$  and  $l_2 : y = \frac{1}{2}x + 3$  are perpendicular and intersect at  $P$ . The three lines  $l_1$ ,  $l_2$  and the  $x$ -axis enclose a triangle  $T$ . What is the area of  $T$ ?

- A)  $\frac{81}{5}$   
 B)  $\frac{54}{5}$   
 C)  $\frac{81}{10}$   
 D)  $\frac{27}{2}$

**Answer:** A)  $\frac{81}{5}$

**Method:** Read the  $x$ -intercepts straight off:  $l_1$  at  $x = 3$ ,  $l_2$  at  $x = -6$ , so the base is 9. The height is the  $y$  of  $P$ :  $6 - 2x = \frac{1}{2}x + 3$  gives  $x = \frac{6}{5}$ ,  $y = \frac{18}{5}$ . Area =  $\frac{1}{2} \cdot 9 \cdot \frac{18}{5} = \frac{81}{5}$ . Option C halves twice; option B uses the wrong base 6.

**Investigate further:** Change  $l_2$  to  $y = \frac{1}{2}x - 3$ . Its  $x$ -intercept moves to 6, and  $P$  drops below the axis. Show the area becomes  $\frac{9}{5}$ , and explain why the triangle now sits under the  $x$ -axis.

2. The perpendicular bisector of the segment joining  $A(2, -6)$  and  $B(5, 4)$  meets the  $x$ -axis at  $P$ . What is the  $x$ -coordinate of  $P$ ?

- A)  $\frac{41}{6}$   
 B)  $\frac{1}{3}$   
 C)  $\frac{19}{5}$   
 D)  $\frac{1}{6}$

**Answer:** D)  $\frac{1}{6}$

**Method:** Skip the gradient:  $P = (p, 0)$  is equidistant from  $A$  and  $B$ , so  $(p - 2)^2 + 36 = (p - 5)^2 + 16$ . The  $p^2$  terms cancel:  $-4p + 40 = -10p + 41$ , so  $p = \frac{1}{6}$ .

**Investigate further:** The equidistance route turns any perpendicular-bisector question into one linear equation. Use it to find where the bisector of  $A(2, -6)$ ,  $B(5, 4)$  meets the  $y$ -axis, and check against the gradient route.

3. The point  $A(2, 9)$  is reflected in the line  $x + 2y = 5$  to give the point  $A'$ . How far is  $A'$  from the origin?

- A)  $\sqrt{85}$   
 B)  $\sqrt{10}$   
 C) 5  
 D)  $\sqrt{185}$

**Answer:** C) 5

**Method:** Move along the normal  $(1, 2)$ . At  $A$ ,  $x + 2y - 5 = 15$ , and each step of  $(1, 2)$  changes it by  $1^2 + 2^2 = 5$ . Reaching the line takes 3 steps back (the foot is  $(-1, 3)$ ); reflecting takes 6:  $A' = (2, 9) - 6(1, 2) = (-4, -3)$ , so  $|OA'| = 5$ . Option B stops at the foot; option A forgets to move at all; option D uses  $+5$  instead of  $-5$  for the constant.

**Investigate further:** Find the point of the line  $x + 2y = 5$  closest to the origin. Use it to explain why  $|OA'|$  and  $|OA|$  differ, even though a reflection preserves distances to every point of the mirror.

4. The lines  $3x - 4y + 2 = 0$  and  $4x - 3y - 5 = 0$  cross, dividing the plane into four angular regions. Which line bisects the angle whose region contains the point  $(1, 1)$ ?

- A)  $x - y = 7$
- B)  $x + y = 7$
- C)  $7x + 7y = 3$
- D)  $7x - 7y = 3$

**Answer:** D)  $7x - 7y = 3$

**Method:** Both normals have length 5, so the bisectors are  $(3x - 4y + 2) = \pm(4x - 3y - 5)$ . At  $(1, 1)$  the two left-hand sides are 1 and  $-4$ : opposite signs. Points in that region satisfy  $3x - 4y + 2 = -(4x - 3y - 5)$ , i.e.  $7x - 7y = 3$ . Option B is the other bisector, the one for the regions where the signs agree.

**Investigate further:** Which of the two bisectors cuts the *acute* angle between the lines? Decide using the sign of the dot product of the normals  $(3, -4)$  and  $(4, -3)$ .

5. The points  $A(1, 4)$ ,  $B(4, 7)$  and  $C(10, k)$  are collinear. The line through  $C$  perpendicular to  $AB$  meets the  $y$ -axis at  $D$ . What is the  $y$ -coordinate of  $D$ ?

- A) 23
- B) 24
- C) 25
- D) 26

**Answer:** A) 23

**Method:**  $AB$  has gradient 1, so  $k = 4 + 9 = 13$ . The perpendicular through  $C(10, 13)$  has gradient  $-1$ :  $y = 23 - x$ , which meets the  $y$ -axis at 23.

**Investigate further:** If  $B$  were  $(4, 10)$  instead,  $AB$  has gradient 2. Show that  $k = 22$  and  $D$  becomes  $(0, 27)$ , then find  $D$  in general when  $AB$  has gradient  $m$ .

6. Triangle  $ABC$  has vertices  $A(0, 0)$ ,  $B(9, 3)$  and  $C(3, 9)$ . The point  $P$  lies on  $AB$  with  $AP : PB = 1 : 2$ , and  $Q$  lies on  $AC$  with  $AQ : QC = 2 : 1$ . What is the area of the quadrilateral  $PBCQ$ ?

- A) 24
- B) 28
- C) 27

D) 32

**Answer:** B) 28

**Method:**  $[ABC] = \frac{1}{2}|9 \cdot 9 - 3 \cdot 3| = 36$ . Triangle  $APQ$  shares angle  $A$ , so  $[APQ] = \frac{1}{3} \cdot \frac{2}{3} \cdot 36 = 8$ , and  $[PBCQ] = 36 - 8 = 28$ . Option D uses  $\frac{1}{3}$  for both ratios; option C treats  $P$  and  $Q$  as midpoints; option A keeps only one ratio.

**Investigate further:** For which ratio  $AQ : QC$  (keeping  $AP : PB = 1 : 2$ ) does  $PBCQ$  have exactly three times the area of  $APQ$ ?

7. The lines  $y = 2x + 1$ ,  $y = 7 - x$  and  $y = \frac{1}{2}x - 2$  enclose a triangle. What is its area?

A) 12

B) 18

C) 24

D) 48

**Answer:** C) 24

**Method:** Pairwise intersections:  $(2, 5)$ ,  $(6, 1)$  and  $(-2, -3)$ . Shoelace:  $\frac{1}{2}|(2-30)+(-18+2)+(-10+6)| = \frac{1}{2} \cdot 48 = 24$ . Option D forgets the  $\frac{1}{2}$ ; option A halves twice.

**Investigate further:** The vertices  $(2, 5)$  and  $(-2, -3)$  lie on  $y = 2x + 1$ . Use that side as the base and the perpendicular distance from  $(6, 1)$  as the height, and check that you get 24 again.

8. The quadrilateral  $Q$  has vertices  $(0, 0)$ ,  $(4, 0)$ ,  $(5, 3)$  and  $(1, 3)$  in that order (a parallelogram). A second quadrilateral  $Q'$  is obtained by reflecting  $Q$  in the line  $y = x$ . What is the area of the overlap  $Q \cap Q'$ ?

A) 6

B) 8

C) 9

D) 12

**Answer:** A) 6

**Method:**  $Q$  is  $0 \leq y \leq 3$ ,  $y \leq x \leq y + 4$ . Reflecting swaps  $x$  and  $y$ , so  $Q'$  is  $0 \leq x \leq 3$ ,  $x \leq y \leq x + 4$ . Both hold on the quadrilateral  $(0, 0)$ ,  $(3, 1)$ ,  $(3, 3)$ ,  $(1, 3)$ : the square  $[0, 3]^2$  (area 9) minus the triangles  $(0, 0)$ ,  $(3, 0)$ ,  $(3, 1)$  and  $(0, 0)$ ,  $(0, 3)$ ,  $(1, 3)$  (each  $\frac{3}{2}$ ), so  $9 - 3 = 6$ . Option C forgets to remove the triangles; option D is the area of  $Q$  itself.

**Investigate further:** Slide  $Q$  one unit to the right, to vertices  $(1, 0)$ ,  $(5, 0)$ ,  $(6, 3)$ ,  $(2, 3)$ . Reflect it in  $y = x$  again and find the new overlap.

## Section D Challenge

(TMUA / SMC / BMO1 difficulty)

*Hardest 5: deep insight, hidden structure, or a construction you must discover. No diagram gives away the trick.*

1. The points  $A(1, 3)$  and  $B(7, 5)$  are fixed, and  $P$  moves along the line  $y = x$ . What is the least possible value of  $PA + PB$ ?

- A)  $4\sqrt{2}$
- B)  $2\sqrt{34}$
- C) 10
- D)  $2\sqrt{10}$

**Answer:** D)  $2\sqrt{10}$

**Method:** Check sides first:  $A$  has  $y > x$  and  $B$  has  $y < x$ , so they are on *opposite* sides of  $y = x$ . The segment  $AB$  crosses the line at  $(4, 4)$ , and  $PA + PB \geq AB = \sqrt{36 + 4} = 2\sqrt{10}$ , with equality there. Option A reflects  $A$  anyway (the reflection trick is only needed when both points are on the same side); option C reflects in the  $x$ -axis; option B reflects through the origin.

**Investigate further:** Move  $B$  to  $(5, 7)$ , on the same side as  $A$ . Now the reflection is needed: find the least value of  $PA + PB$  and the point  $P$  that achieves it.

2. The square  $MNOP$  has side 10. Points  $R, S, T, U$  lie on  $MN, NO, OP, PM$  respectively, with  $MR = x$ , and  $RSTU$  is a rectangle whose sides make  $45^\circ$  with the sides of the square. For  $0 < x < 10$ , what is the largest value of  $x$  for which the area of  $RSTU$  is 20?

- A)  $5 + \sqrt{5}$
- B)  $5 + \sqrt{15}$
- C)  $5 + 2\sqrt{5}$
- D)  $10 - \sqrt{5}$

**Answer:** B)  $5 + \sqrt{15}$

**Method:** Put  $M = (0, 0)$ ,  $R = (x, 0)$ . The  $45^\circ$  sides force  $S = (10, 10 - x)$ ,  $T = (10 - x, 10)$  and  $U = (0, x)$ . Then  $RS = \sqrt{2}(10 - x)$  and  $RU = \sqrt{2}x$ , so the area is  $2x(10 - x)$ . Setting  $2x(10 - x) = 20$  gives  $x^2 - 10x + 10 = 0$ , so  $x = 5 \pm \sqrt{15}$ , and the larger is  $5 + \sqrt{15}$ . Option A drops the factor 2 in the area; option C solves for area 10.

**Investigate further:** For which  $x$  is  $RSTU$  a square, and what is the largest area such a rectangle can have? Explain why both answers come from the same  $x$ .

3. What is the area of the region of points  $(x, y)$  satisfying both  $|x| + 2|y| \leq 6$  and  $y \geq x - 2$ ?

- A) 10
- B) 18
- C) 26
- D) 36

**Answer:** C) 26

**Method:**  $|x| + 2|y| \leq 6$  is the rhombus with vertices  $(\pm 6, 0)$ ,  $(0, \pm 3)$ , of area  $\frac{1}{2} \cdot 12 \cdot 6 = 36$ . The line  $y = x - 2$  cuts its edges at  $(-\frac{2}{3}, -\frac{8}{3})$  and  $(\frac{10}{3}, \frac{4}{3})$ . The part *below* the line is the quadrilateral  $(-\frac{2}{3}, -\frac{8}{3}), (0, -3), (6, 0), (\frac{10}{3}, \frac{4}{3})$ , with shoelace area 10. So the required area is  $36 - 10 = 26$ . Option A is the piece cut off; option B assumes the line halves the rhombus.

**Investigate further:** For which  $c$  does the line  $y = x + c$  split the rhombus into two equal areas? Explain why symmetry forces the answer.

4. How many straight lines through the point  $(2, 3)$  form, together with the two coordinate axes, a triangle of area 12?

A) 1  
B) 2  
C) 4  
D) 3

**Answer:** D) 3

**Method:** A line of gradient  $m$  through  $(2, 3)$  has intercepts  $2 - \frac{3}{m}$  and  $3 - 2m$ , so the area is  $\frac{(2m-3)^2}{2|m|}$ . Setting it to 12: for  $m > 0$ ,  $4m^2 - 36m + 9 = 0$  has two positive roots; for  $m < 0$ ,  $4m^2 + 12m + 9 = (2m+3)^2 = 0$  gives  $m = -\frac{3}{2}$  once. That is 3 lines. Option B misses the negative branch; option C counts the double root twice.

**Investigate further:** The value 12 is exactly the *smallest* area of a first-quadrant triangle through  $(2, 3)$ , which is why the negative branch has a double root. Prove that the minimum occurs when  $(2, 3)$  is the midpoint of the hypotenuse.

5. For each  $t$  with  $0 < t < 4$ , the line  $L_t$  passes through  $(t, 0)$  and  $(0, 4 - t)$ . What is the greatest possible distance from the origin to  $L_t$ ?

A) 1  
B) 2  
C)  $\sqrt{2}$   
D)  $2\sqrt{2}$

**Answer:** C)  $\sqrt{2}$

**Method:** With  $a = t$ ,  $b = 4 - t$  the distance is  $d = \frac{ab}{\sqrt{a^2 + b^2}}$ , and  $a^2 + b^2 = 16 - 2ab$ . So  $d^2 = \frac{p^2}{16 - 2p}$  with  $p = ab$ , which increases with  $p$ . The product is largest ( $p = 4$ ) at  $a = b = 2$ , giving  $d^2 = 2$  and  $d = \sqrt{2}$ . Option B reports  $d^2$ ; option D is the distance from the origin to the midpoint  $(2, 2)$  of the segment, not to the line.

**Investigate further:** Every  $L_t$  is tangent to one fixed curve, the envelope of the family. Find it, and check that the foot of the perpendicular from the origin to  $L_2$  lies on it.

## Top 5 Patterns — Worksheet #1

- Implicit line  $\rightarrow$  gradient  $-a/b$ .** Read it off  $ax + by + c = 0$  without rearranging; the constant never touches the gradient. (see A1, A7, B7)
- Perpendicular bisector = equidistance.**  $PA^2 = PB^2$  is one linear equation and needs no gradient at all. (see B3, C2, C5)
- Move along the normal.** Distance to a line, the foot of a perpendicular and a reflection all come from stepping along  $(a, b)$  from the point. (see A6, B2, C3, C4)
- Shoelace in cyclic order.** Half the absolute cross-term sum; vertices must go round the shape in order. (see A10, B4, B10, C7, D3)

- 5. Area by ratio before area by coordinates.** Intercepts, shared angles and ratios give most areas in one line. *(see C1, C6, D2, D4)*

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**Common Traps — Worksheet #1**

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- 1. Half a perpendicular flip.**  $m = \frac{2}{3}$  pairs with  $-\frac{3}{2}$ : invert *and* negate. *(see A3, B1)*
- 2. Stopping at the foot.** A reflection travels twice as far along the normal as the foot of the perpendicular. *(see C3)*
- 3. Picking the wrong bisector.** Two lines have two angle bisectors; the signs of the line expressions at a test point decide which one you want. *(see C4)*
- 4. Reflecting when you don't need to.** If the two points are on opposite sides of the line, the straight segment already crosses it. *(see D1)*
- 5. Losing a branch.** Absolute values and “largest  $x$ ” questions hide extra solutions: check every sign case before counting. *(see D2, D4)*

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*“Shoelace before altitude: area is a one-line cross product, not a construction.”*

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Found a mistake or want to contribute a sheet?  
Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)